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OPERATING SYSTEM CONCEPTS LAB EXPERIMENTS

EXP – 05: SCHEDULING ALGORITHMS (First Come First Serve)

Activities Terminal Mar 25 00:07

user123@localhost:~/scheduling

```
[user123@localhost ~]$ ls
abc.sh  Desktop    Downloads  ifelse.sh  Music  Pictures   Public   REC       sample.sh   SFA  Templates  while.sh
ab.sh   Documents  Folder     loops.sh   pc     posneg.sh  qns.txt  sample.c  sample.txt  sh   Videos
[user123@localhost ~]$ mkdir scheduling
[user123@localhost ~]$ cd scheduling
[user123@localhost scheduling]$ vi fcfs.c
[user123@localhost scheduling]$ cat fcfs.c
#include <stdio.h>

int main(){
        int n;
        printf("Enter number of processes: ");
        scanf("%d",&n);

        int at[n], bt[n], ct[n], tat[n], wt[n];

        for(int i=0; i<n; i++){
                printf("Enter Arrival time and Burst Time for P%d: ", i+1);
                scanf("%d %d:",&at[i],&bt[i]);
        }

        float total_tat = 0, total_wt = 0;

        ct[0] = at[0] + bt[0];

        for(int i=0; i<n; i++){
                if(ct[i-1]<at[i]){
                        ct[i] = at[i] + bt[i];
                } else {
                        ct[i] = ct[i-1] + bt[i];
                }
        }

        for(int i=0; i<n; i++){
                tat[i] = ct[i] - at[i];
                wt[i] = tat[i] - bt[i];

                total_tat += tat[i];
                total_wt += wt[i];
        }

        printf("\nAverage Turnaround Time: %.2f", total_tat/n);
        printf("\nAverage Waiting Time: %.2f", total_wt/n);

        return 0;
}
[user123@localhost scheduling]$ gcc fcfs.c
[user123@localhost scheduling]$ ./a.out
Enter number of processes: 4
Enter Arrival time and Burst Time for P1: 0 2
Enter Arrival time and Burst Time for P2: 1 2
Enter Arrival time and Burst Time for P3: 5 3
Enter Arrival time and Burst Time for P4: 6 4

Average Turnaround Time: 3.50
Average Waiting Time: 0.75[user123@localhost scheduling]$
```